Version 1 Basic Bayesian model

$y = |x - \mu| \times \mu$

GPD

$$f(\sigma, \xi, y) = \begin{cases} \frac{1}{\sigma}\left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi - 1} & , \xi \neq 0 \\ \frac{1}{\sigma} e^{-y/\sigma} & , \xi = 0 \end{cases}$$

take log for numerical stability

$$\log(f(\sigma, \xi)) \begin{cases} \xi = 0: \sum_{n=1}^{N} \left[-\log(\sigma) - \frac{y_n}{\sigma}\right] \\ \xi \neq 0: \sum_{n=1}^{N} \left[-\log(\sigma) - \left(\frac{1}{\xi} + 1\right)\log\left(1 + \frac{\xi y_n}{\sigma}\right)\right] \end{cases}$$

(σ) prior: $\sigma > 0$ for $\pi(\sigma) \propto \underbrace{\sigma^{2-1} e^{-.1\sigma}}_{\text{comes from } \sigma \sim \text{Gamma}(\alpha = 2, \beta =}$

(ξ) prior: $\xi \sim \text{Normal}(\mu = 0, \sigma = .3) \Rightarrow [-.3, .3]$

prior σ

$$\pi(\xi) \propto \frac{1}{\tau\sqrt{2\pi}} e^{\left(\frac{(\xi - \mu)^2}{2\tau^2}\right)}$$

$$\Leftrightarrow \pi \propto e^{\left(\frac{-\xi^2}{.18}\right)} \text{ for } -.3 < \xi < 1$$

likelihoods

$$L(\sigma \mid y) = \frac{1}{\sigma} e^{\left(-\frac{y}{\sigma}\right)}, \quad \varepsilon = 0$$

$$L(\sigma, \varepsilon \mid y) = \frac{1}{\sigma}\left(1 + \frac{\varepsilon y}{\sigma}\right)^{\frac{1}{\varepsilon} - 1}$$

$\log(L(\ldots))$

$\hookrightarrow \varepsilon = 0: \sum_{n=1}^{N} \left[-\log(\sigma) - \frac{y_n}{\sigma}\right]$ ✓

$\hookrightarrow \varepsilon \neq 0: \sum_{n=1}^{N} \left[-\log(\sigma) - \left(\frac{1}{\varepsilon} + 1\right)\log\left(1 + \frac{\varepsilon y_n}{\sigma}\right)\right]$ ✓

$\log(f(\sigma, \varepsilon, y)) = \log(\text{likelihood})$

data = $N \geq 1$:= # exceedances

$[N]\ x > 0$:= exceedance value for each exceedance #

$\sigma \sim \text{Gamma}(\alpha = 2, \beta = .1)$

$\varepsilon \sim \text{Normal}(\mu = 0, \sigma = .3)$ => informed by physical intuition

rasthfeed lab

likelihood / gpd_log / ~~gpd_lpdf~~

gpd_lpdf (x, σ, ε)

$\hookrightarrow \varepsilon = 0: -\log(\sigma) - \frac{x}{\sigma}$

$\hookrightarrow \varepsilon \neq 0: -\log(\sigma) - \left(\frac{1}{\varepsilon + 1}\right)\log\left(1 + \frac{\varepsilon x}{\sigma}\right)$

$\log(1 + \frac{\varepsilon x}{\sigma})$